

Marcus Benett Zaens

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EDUCATION

Plano East Senior High School (Plano, Texas) <i>International Baccalaureate Program, Grades 9-12</i>	2023 - 2027
St. Mark Catholic School (Plano, Texas) <i>Honors Program, Grades 5-8</i>	2019 - 2023
St. Catherine of Siena School (Manchester, New Hampshire) <i>Grades 1-4</i>	2015 - 2019

Relevant Coursework

Computer Science: IB HL Computer Science, AP Computer Science Principles

Math: IB HL Math Analysis & Approaches, AP Calculus BC, AP Precalculus, Algebra 1 & 2 Honors, Geometry Honors

Physics: IB SL Physics, PLTW Engineering Science

PROJECTS

Personal Portfolio

- Developing a portfolio website using JavaScript, HTML, and CSS to showcase projects and technical skills
- Managing version control using Git and hosted the project in a Github repository for efficient updates and deployment

Robotic Arm

- Engineered a robotic arm using an Arduino Uno Rev3 microcontroller, servo motors, and potentiometers
- Designed and fabricated custom mechanical parts in Fusion 360 using 3D printing
- Programmed motor control and motion sequences in C, achieving reliable and repeatable movements

Custom Mechanical Keyboard (<https://github.com/mbzgr8/keyboard-firmware>)

- Designed and routed a custom PCB in KiCAD, implementing a diode-protected key matrix
- Soldered diodes, mechanical switches, and a RP2040 Pro Micro microcontroller
- Programmed an RP2040 Pro Micro to support QMK firmware and VIA compatibility for custom keymaps
- Created a custom 4-part keyboard case in Fusion 360

Pygame Chicken Catcher Game

- Created an arcade-style endless catcher minigame where you catch eggs, dodge falling bombs, and eat spicy peppers
- Built with Python's Pygame library: including features like collision detection, animations, and sound effects
- Includes score tracking, high score saving, difficulty scaling, countdown timer, and a homescreen
- Designed pixel-art sprites and images for animation and a better game experience

Predicting House Prices with Regression & Neural Networks

- Used the Ames Housing dataset to predict house prices using pandas, numpy, tensorflow, and scikit-learn
- Built linear, multiple, and polynomial regression models as well as a neural network
- Implemented a 30% train and test split for regression and polynomial features for a better fit
- Performed exploratory data analysis using matplotlib with scatter plots, histograms, and bar graphs
- Evaluated models using MSE to determine margin of error

EXTRACURRICULARS

- Math Club
- Chess Club
- Computer Science Club
- Environmental Safety Club
- Mu Alpha Theta
- US Chess Federation
- Piano Student
- FIRST Lego League (2018 - 2019)

ACHIEVEMENTS AND AWARDS

- Plano Scholar (2023 - Present)
- National Honor Society Member (2024 - Present)
- AP Scholar with Honor/Distinction (2025/2026)
- Speech and Debate Competitions
 - Newman High School, February 2023
 - Plano West High School, October 2022
 - McMillen High School, September 2022
- FIRST Lego League International Open Competition 2019

LANGUAGES, SKILLS AND CERTIFICATES

Languages: English, Spanish, Tagalog

Programming Languages: Python, Java, C, Javascript, HTML/CSS

Software Development Tools: Git, Github, Visual Studio Code, Linux

Hardware Development Tools

- Arduino Uno, Breadboarding, Soldering
- *CAD and Design Tools:* Fusion 360, Blender, KiCAD

Certificates: IBM Artificial Intelligence Fundamentals, NOCTI Engineering Technology Foundations

WORK EXPERIENCE

Math and Reading Student Instructor, Kumon

June 2023 - Present

- Tutored groups of 3-4 students in math and reading, improving comprehension, problem-solving skills, and confidence
- Graded homework, administered classwork, and managed student records
- Monitored student progress and adjusted lesson plans to ensure continuous improvement

COMMUNITY SERVICE AND VOLUNTEER WORK

- International Baccalaureate Mentorship Program
- Love for Our Elders
- Distinguished Member in Environmental Safety Club

INTERESTS

- Mathematics
- Programming
- Computer Systems and Hardware
- Embedded Systems
- 3D Modeling
- Artificial Intelligence and Machine Learning

HOBBIES

- Chess
- Playing the Piano
- Video Games
- Solving Puzzles